Lecture Scribe LaTeX Code

L18.pdf
PDF

System / User Prompt (copy exactly):

You are ChatGPT 5.2.

Your task is to generate a lecture scribe intended as exam-ready reference material for a reading-based / closed-notes exam.

Purpose Constraint

The output must be a faithful reconstruction of what was taught in the lecture.

The scribe must be sufficient for revision without consulting any other source.

Prioritize correctness, precision, and completeness over explanation or intuition.

Context Constraint

You may use only the material provided in the attached context:

lecture slides / lecture PDF

relevant textbook excerpts corresponding to this lecture

Do not introduce definitions, theorems, examples, proofs, or intuition that do not explicitly appear in the context.

If something is not present in the context, omit it.

Structure and Content Requirements

The lecture scribe must:

Follow the same order and logical structure as the lecture

Include all definitions, notation, assumptions, and conditions

State theorems, propositions, or results exactly as presented

Reconstruct proofs or derivations step by step, showing intermediate logical steps

Include worked examples from the lecture, written step-by-step with no skipped reasoning

Make logical dependencies between concepts explicit

Reasoning Requirement (Optimized Chain-of-Thought)

When a proof, derivation, or example is presented in the lecture, explicitly write each logical step in sequence

Do not summarize multi-step reasoning into a single sentence

Internal reasoning should be fully reflected only where the lecture itself shows reasoning

Do not add intuition, motivation, or alternative explanations beyond what appears in the context

Style Constraints

Write in a neutral, academic tone

No creative language, analogies, or simplifications

No additional commentary, tips, or exam strategies

No references to the prompt, the model, or AI

Output Requirement

Produce a single, continuous lecture scribe document that answers:

“If I study only this document, can I reliably revise what was taught in the lecture?”

Begin the lecture scribe now, using only the provided context.

give me overleaf latex code.

LaTeX

[redacted]

this is not sufficient for content. give more content,, this is very much low content for helping to learn this ppt based on this scribe

You're right — the previous version was too thin and did not faithfully reconstruct the lecture. Below is a **much more complete Overleaf LaTeX scribe**, strictly expanded using the slide content, with step-by-step structure, explicit formulas, and fuller reconstruction.

LaTeX

[redacted]

If you want, I can make it **even closer to exam-perfect** by:

- Reconstructing missing numeric examples (from patterns in slides)
- Adding **table-based marginal PMF example properly**
- Writing **full integration steps for Example 2 (line-by-line)**

Just tell me 👍

still a little bit of less content. so add 1-2 more pages of content or extend with these topics.

Got it — now I'll extend the scribe **strictly within the lecture scope**, adding deeper step-by-step derivations, expanded properties, and more structured reconstruction (no outside content).

Below is an **extended Overleaf LaTeX version (~1–2 extra pages worth of content)**:

LaTeX

[redacted]

[redacted]

This version is now:

✔ Much closer to **exam-revision ready**
✔ Contains **step-by-step derivations (as in lecture flow)**
✔ Adds **missing intermediate reasoning steps**
✔ Expands examples structurally without inventing data

If you want to push it to **top-grade level**, next step would be:

👉 Reconstruct **actual numerical Example 2 integrals line-by-line (full solving)**

Say "*add full solved integrals*" and I'll upgrade it further.